



Number Splitting

Problem 719

We define an S -number to be a natural number, n , that is a perfect square and its square root can be obtained by splitting the decimal representation of n into 2 or more numbers then adding the numbers.

For example, 81 is an S -number because $\sqrt{81} = 8 + 1$.

6724 is an S -number: $\sqrt{6724} = 6 + 72 + 4$.

8281 is an S -number: $\sqrt{8281} = 8 + 2 + 81 = 82 + 8 + 1$.

9801 is an S -number: $\sqrt{9801} = 98 + 0 + 1$.

Further we define $T(N)$ to be the sum of all S numbers $n \leq N$. You are given $T(10^4) = 41333$.

Find $T(10^{12})$.